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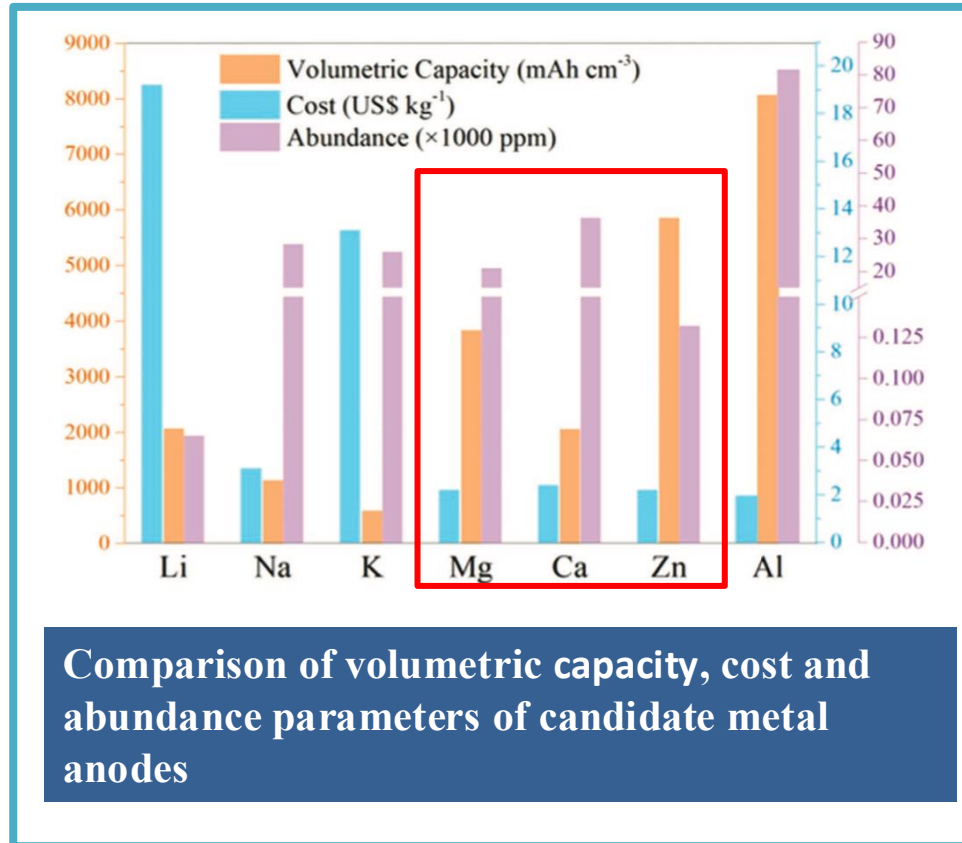
Electrode/Electrolyte Interface Control for Divalent-Ion Batteries with High Stability and Reversibility

Chang Li

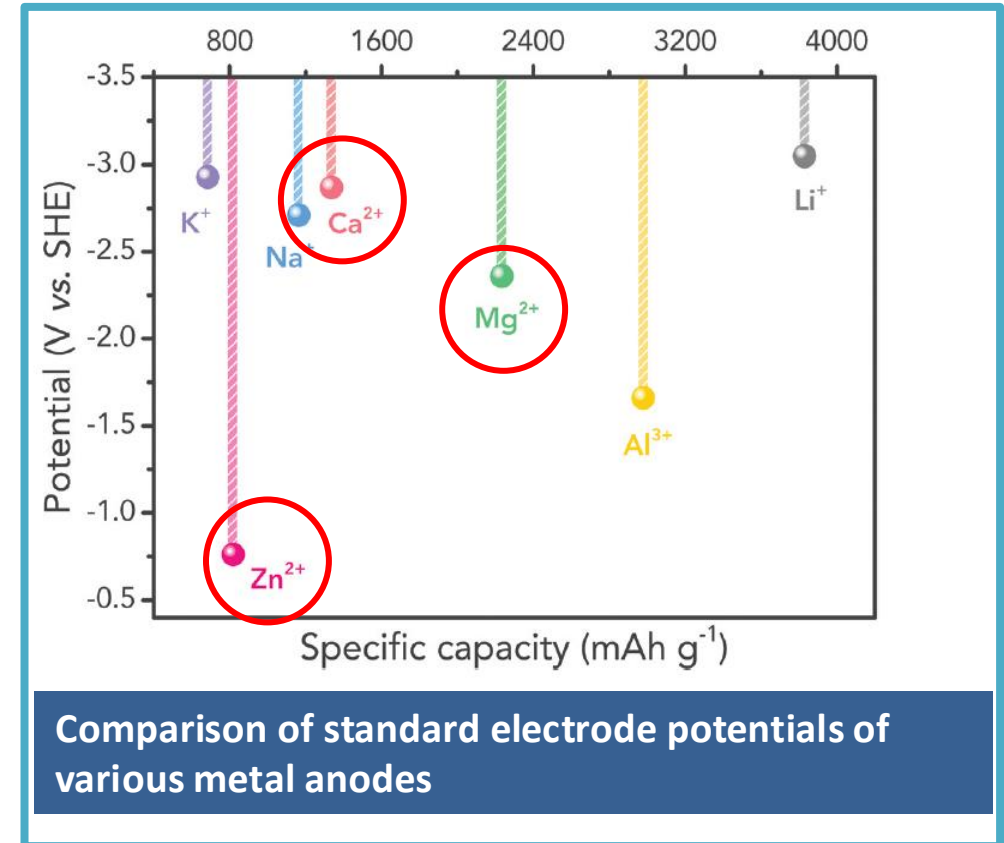
Supervisor: Linda F. Nazar



Divalent-ion batteries: low cost + high volumetric capacity



M. Srinivasan *et al.* *Adv. Sustainable Syst.* **2018**, 3, 1800111.



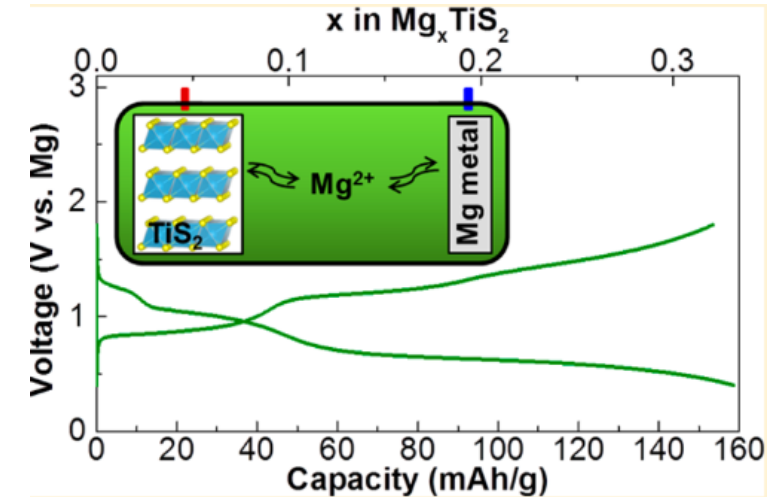
C. Zhi *et al.* *Chemical Society Reviews* **2020**, 49, 180-232

- This talk will focus on achieving **high-voltage Mg batteries** by **controlling the Mg-electrolyte interface**

“Soft” anion cathode + chloride-based electrolyte

Layered TiS_2 ($\sim 160 \text{ mAh g}^{-1}$)

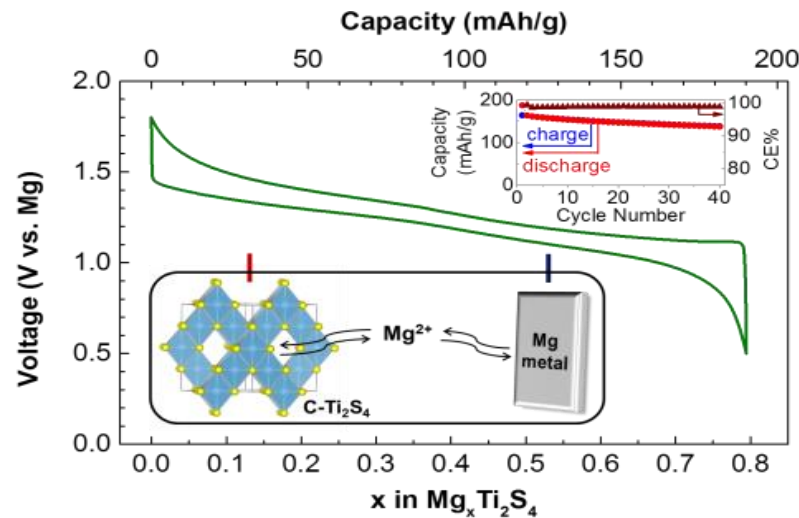
“APC”- $\text{PhMgCl}_2/\text{AlCl}_3$ in THF



L.F. Nazar *et al*, *ACS Energy Lett.* **2016**, 1, 29.

Thiospinel Ti_2S_4 ($\sim 190 \text{ mAh g}^{-1}$)

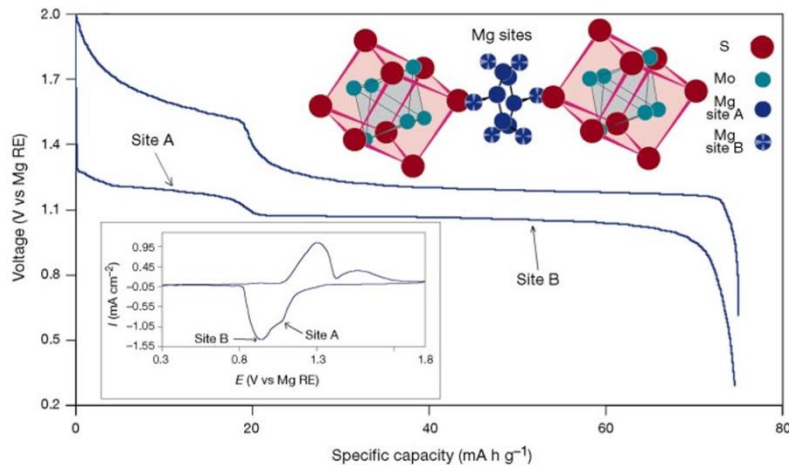
“APC”- $\text{PhMgCl}_2/\text{AlCl}_3$ in THF



L.F. Nazar *et al.*, *Energy Environ Sci.* **2016**, 9, 2273.

Chevrel Mo_6S_8 ($\sim 75 \text{ mAh g}^{-1}$)

0.25 M $\text{Mg}(\text{AlCl}_2\text{BuEt})_2$ in tetrahydrofuran (THF)



D. Aurbach *et al.*, *Nature* **2000**, 402, 724

- Low operation voltage of $< 2 \text{ V}$

The state-of-art Mg-compatible electrolytes

Table 1. Electrochemical properties of key Mg-compatible electrolytes based on classical (chloride-based) and single-salt designs

Electrolyte salts		Solvent	Oxidative stability (V)		Coulombic efficiency (%)	Ionic conductivity at 25°C (mS cm ⁻¹)
			Pt	Stainless steel		
Chloride-based	PhMgCl/AlCl ₃ (APC)	THF	>3 ²⁰	2.1 ³⁰	100 ^{a,20}	≈ 4.3 ²⁰
	MgCl ₂ /AlCl ₃ (MACC)	THF	3.0 ^{30,31} 3.1 ²⁴ 3.4 ²²	2.0 ³¹ 2.1 ³⁰	≈ 90 ^{a,22} 100 ^{a,24,31} 98.8 ^{b,22}	0.23 ²⁴
		DME	3.3 ³¹ 3.4 ³²	2.2 ³¹	95 ^{a,32} 100 ^{a,31}	
		diglyme	3.4 ³¹	2.1 ³¹	94 ^{a,31}	
	MgTFSI ₂ /MgCl ₂	DME	3.3 ³⁰ 3.5 ³²	2.1 ³⁰ 2.4 ²⁸	80 ^{a,32} 93 ^{a,28} ≈ 100 ^{a,26} 94 ^{b,26}	
Chloride-free	Mg(BH ₄) ₂ /LiBH ₄	DME	1.7 ³⁶	2.2 ³⁶	94 ^{a,36}	2.07 ³⁷ 2.1 ³⁸
		THF			≈ 85 ^{a,37}	2.61 ³⁷
		diglyme			100 ^{a,37}	3.27 ³⁷
	Mg(CB ₁₁ H ₁₂) ₂	tetraglyme	3.5 ³⁰ ≥ 3.5 ^{d,39}	3.0 ³⁰ ≥ 3.5 ^{d,39}	>99 ^{a,39}	1.8 ³⁹
		triglyme	≥ 3.5 ^d			2.9 ³⁹
	Mg[B(OCH(CF ₃) ₂) ₄] ₂	DME	≥ 3.5 ^{d,40}	≥ 3.5 ^{d,40}	>98% ^{c,40}	11 ⁴¹
	Mg[B(O ₂ C ₂ (CF ₃) ₄) ₂] ₂	diglyme	≥ 3.5 ^{d,42}	≥ 3.5 ^{d,42}	95 ^{a,42}	3.95 ⁴²
	Mg[Al(OCH(CF ₃) ₂) ₄] ₂	DME	≥ 3.5 ^{d,43}	≥ 3.5 ^{d,43}	>99 ^{a,43}	6.5 ⁴³

^aCE measured by cycling voltammetry.

^bAverage CE measured galvanostatically.

^cCE measured by galvanostatic cycling.

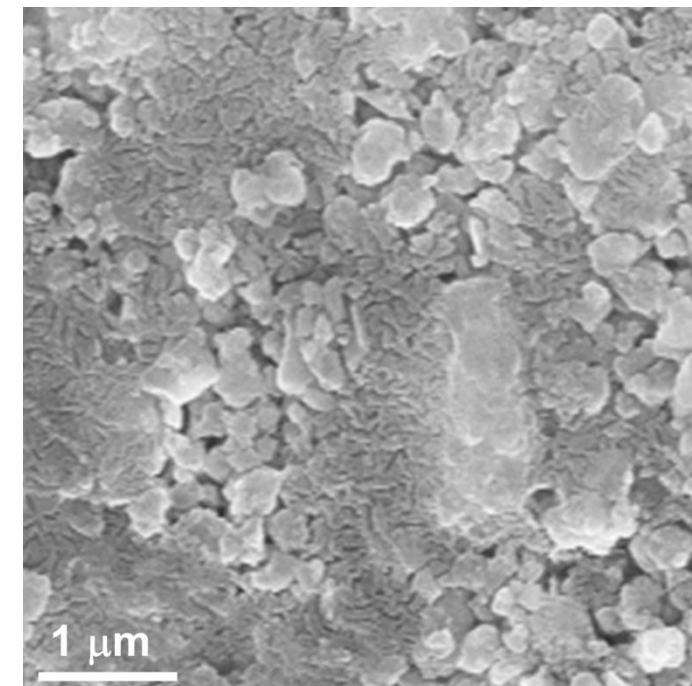
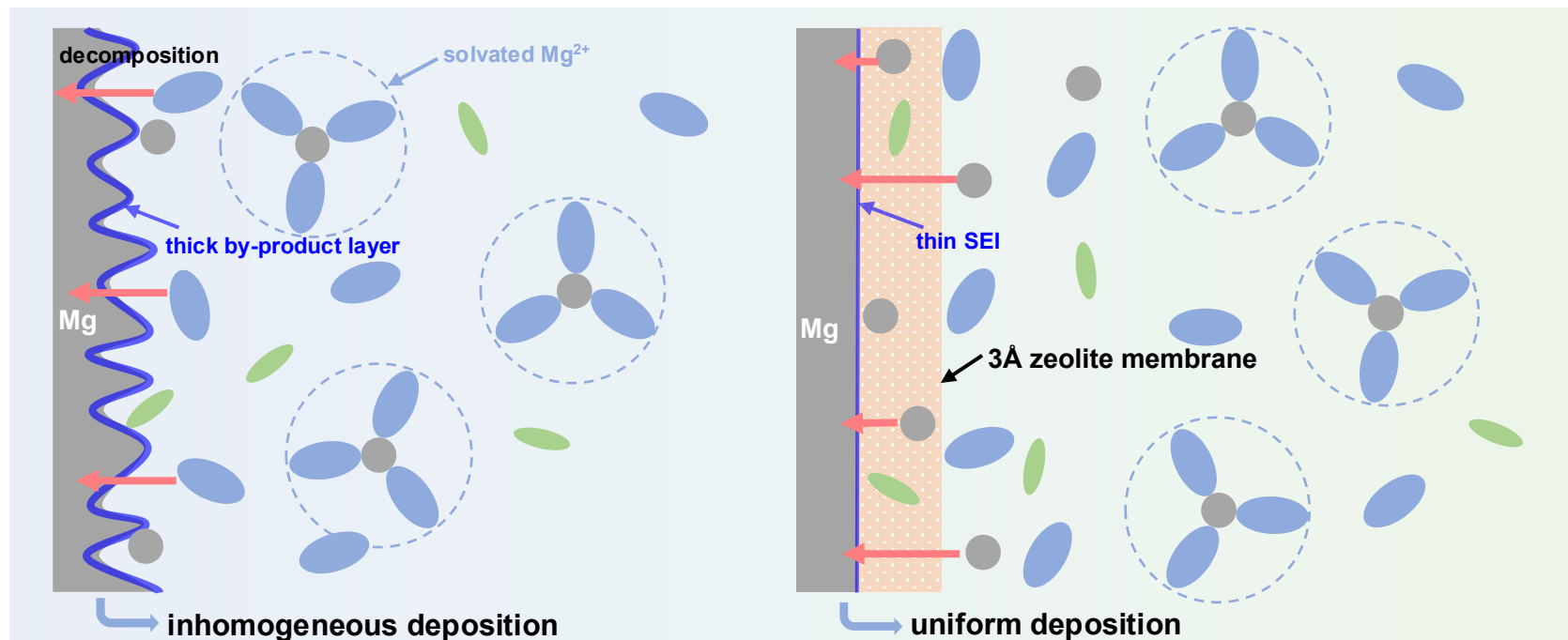
^dReported as limited by ethereal solvent.

- Low oxidative stability
- High corrosivity to battery components

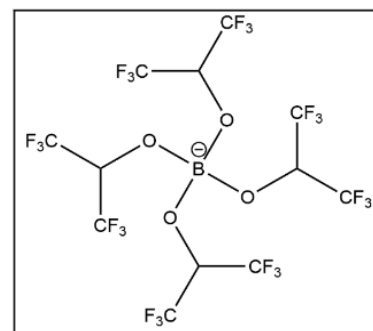
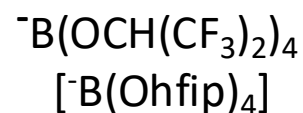
- Commercial unavailability
- Poor Mg anode stability
- Electrolyte conditioning
- Lack of high-voltage cathode

R. Mohtadi *et al.* *Joule* **2021**, 5, 1–37

Inorganic zeolite 3A membrane protects Mg against electrolyte decomposition



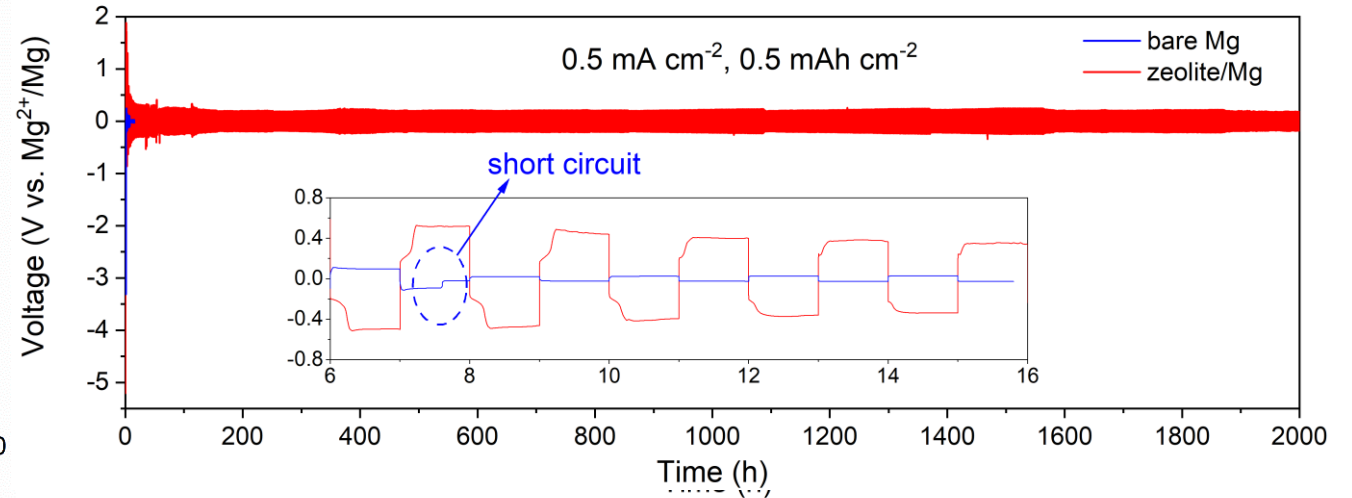
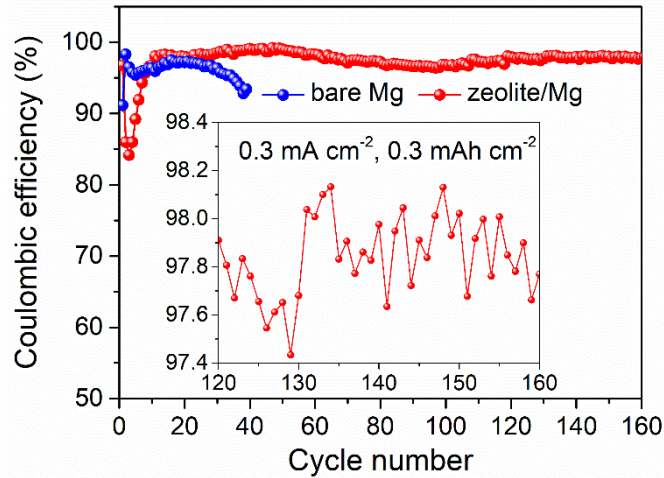
Electrolyte: 0.25 M $\text{MgB}(\text{Ohfip})_4)_2$ / diglyme (G2)



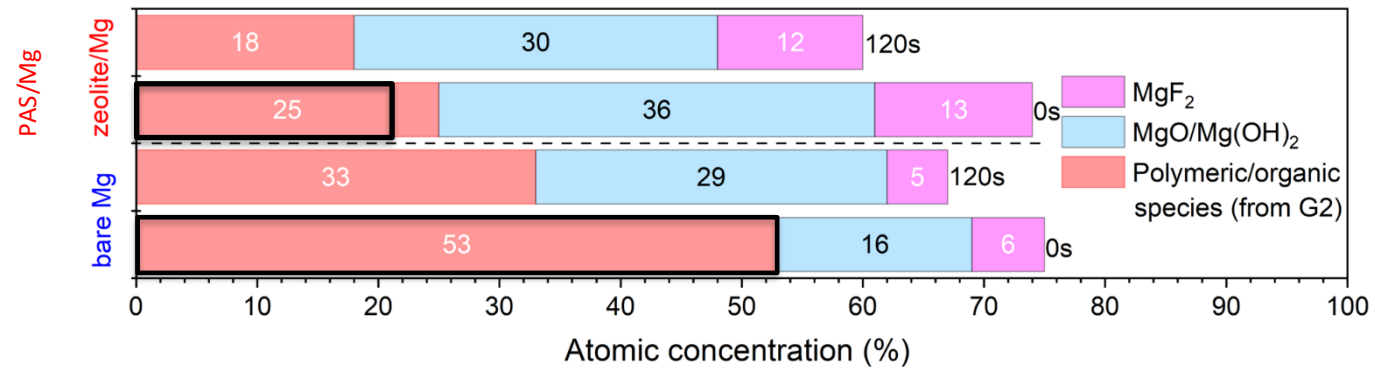
- ~ 150 nm zeolite 3A/PTFE (9:1)
- thickness: ~ 100 μm

Enhanced Mg plating/stripping stability at high currents

- **Bare Mg:**
 - rapid shorting
- **zeolite/Mg:**
 - Stable Mg plating/stripping

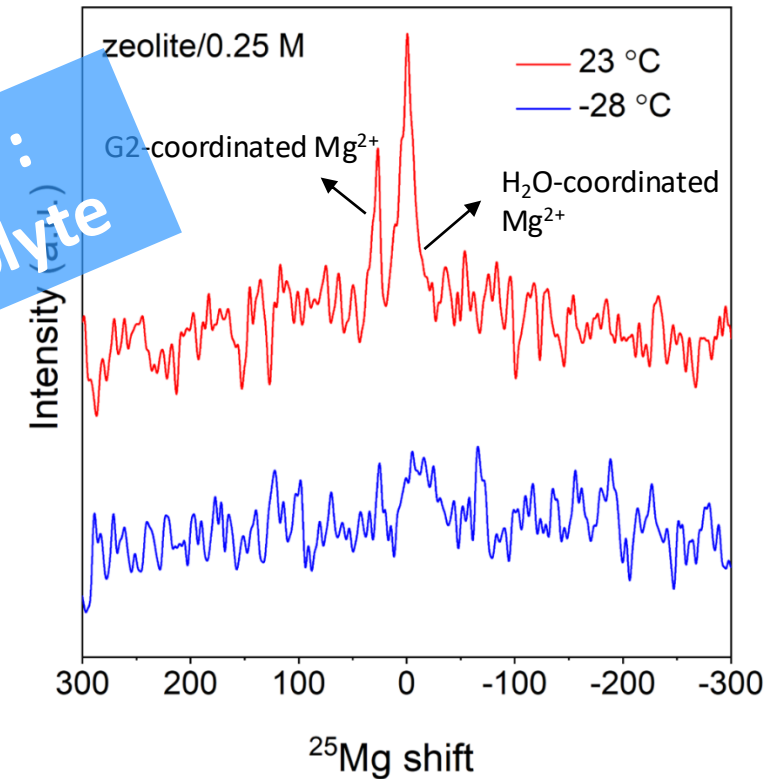
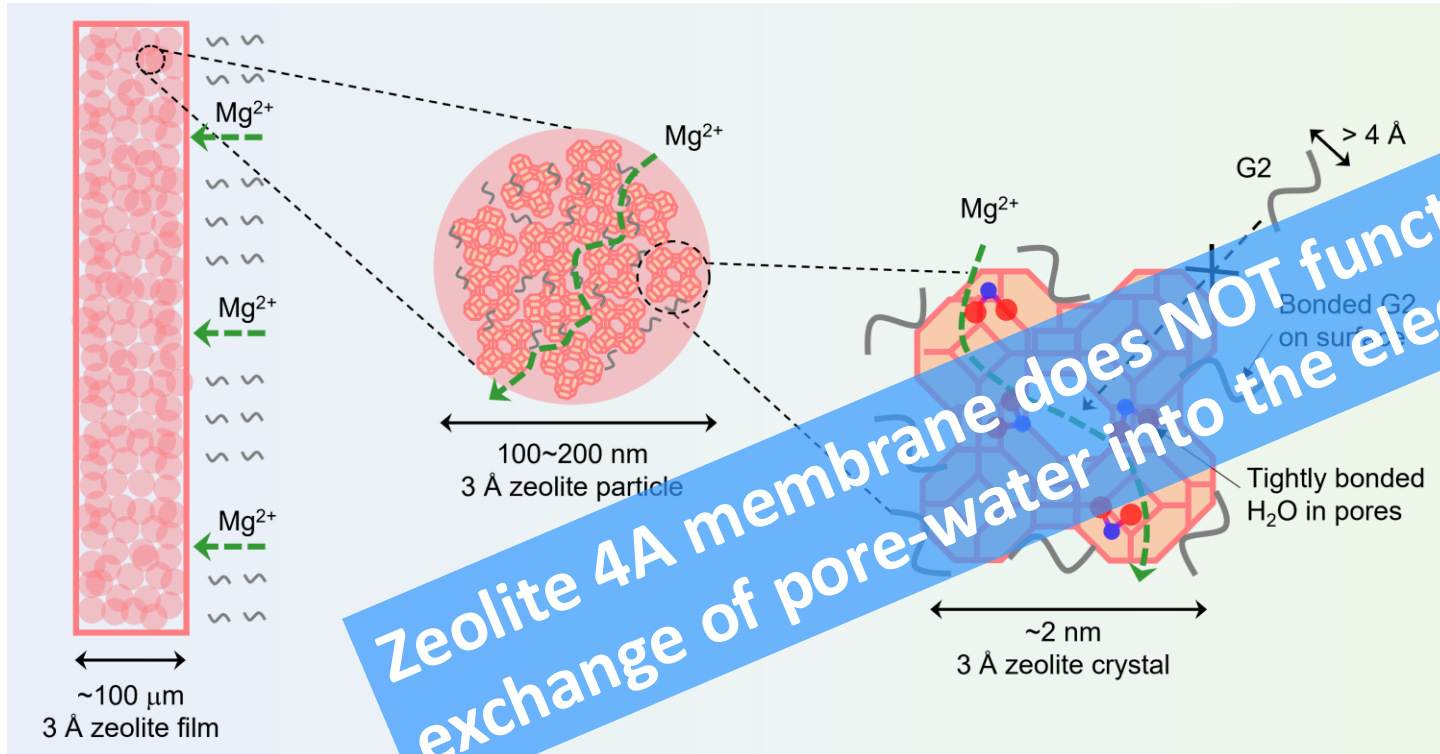


- XPS: homogeneous Mg deposition under zeolite membrane that protects Mg from degrading free G2.

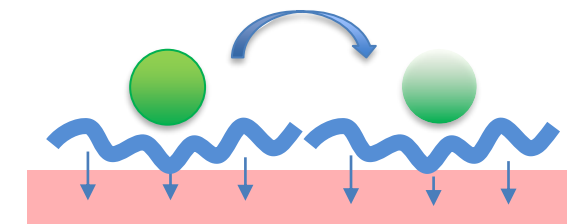


Surface “trapped” free G2 + facile Mg^{2+} migration

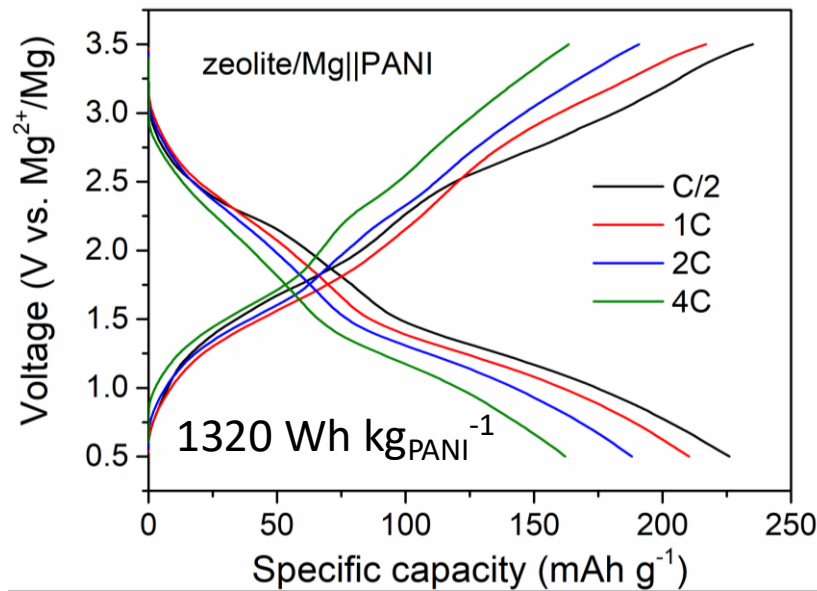
- Two distinct Mg^{2+} migration pathways



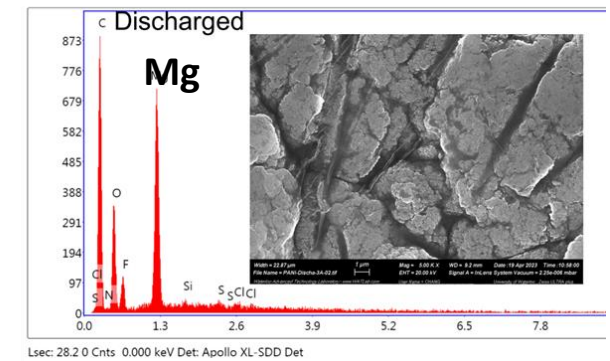
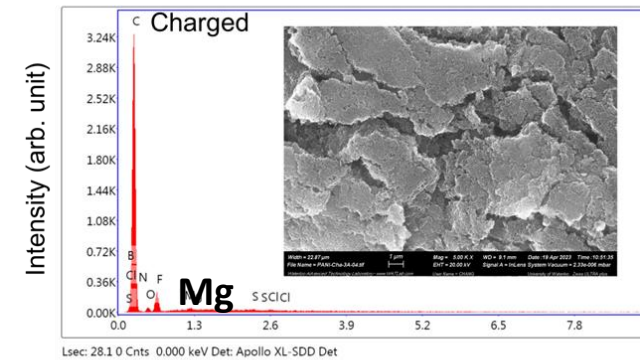
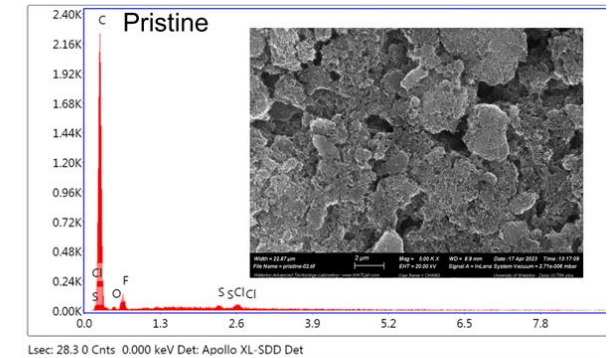
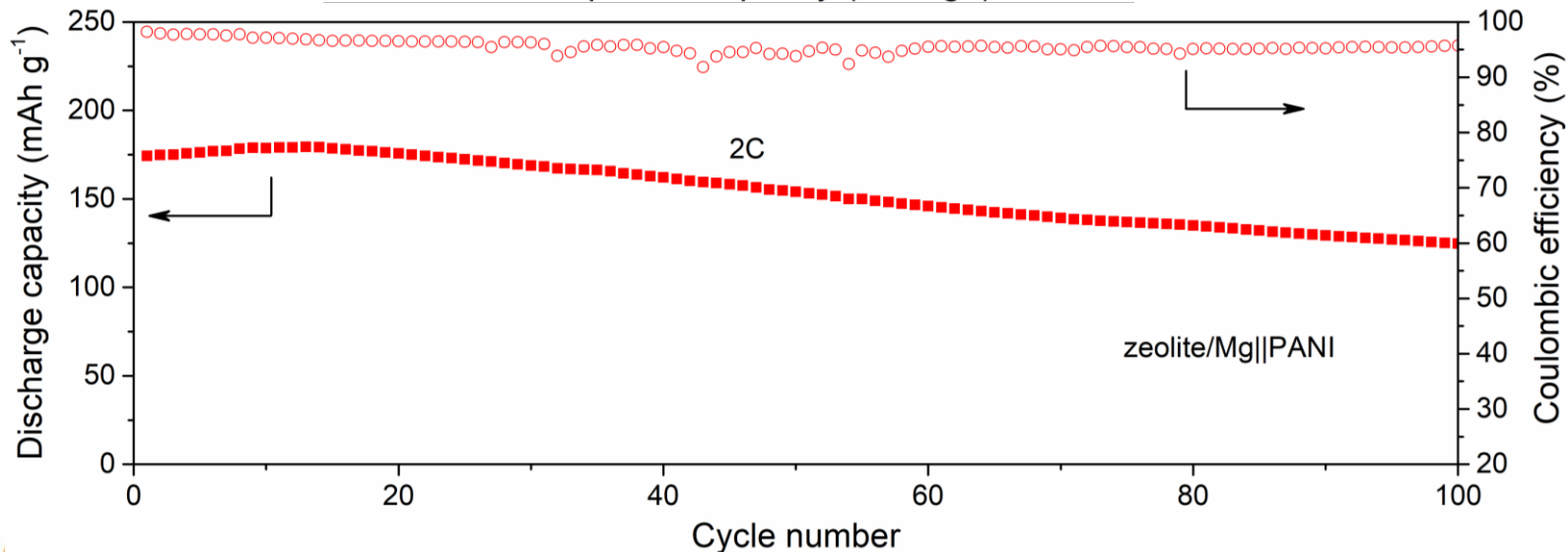
- diglyme strongly adsorbs on zeolite surface:
 - Under-coordinated diglyme – weakly bound Mg^{2+} helps mobility
- rapid ion-exchange between Mg^{2+} and cations in the zeolite:
 - Mg^{2+} cations hop along H_2O molecules in the cages



3.5 V high power zeolite/Mg || organic cathode (PANI) cell



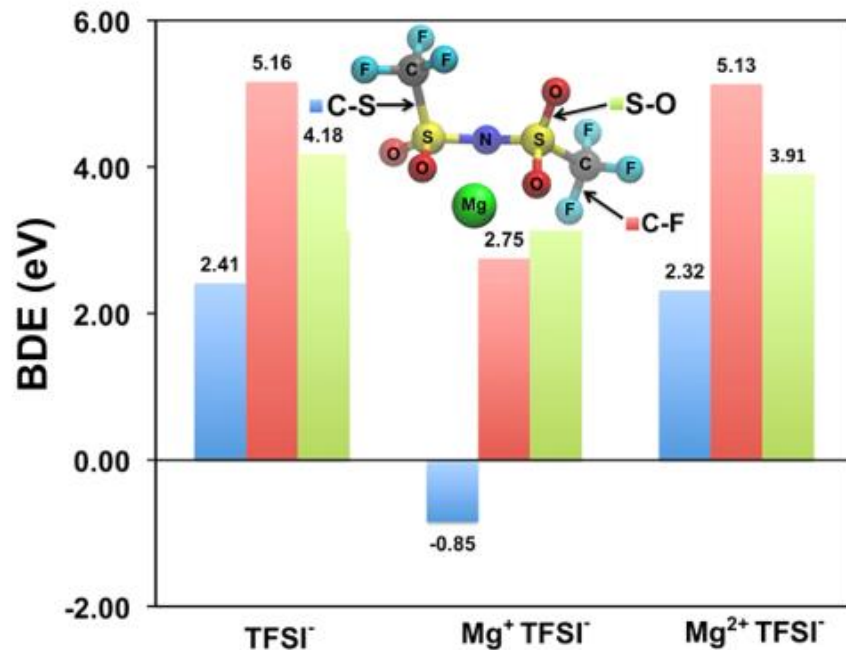
1C = 250 mA g_{PANI}⁻¹
AM: 2.5 mg cm⁻²



- Dominant Mg²⁺ (de)intercalation

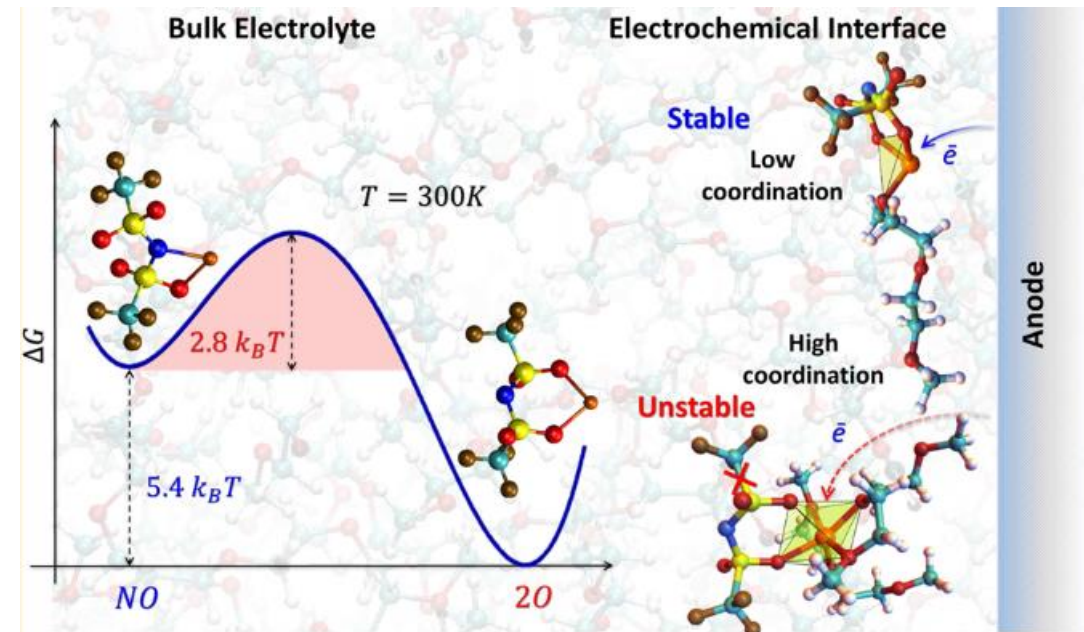
Design of easily scalable electrolyte for high-rate Mg batteries

- Ion-pairing induces difficulty in Mg plating/stripping in conventional $\text{Mg}(\text{TFSI})_2/\text{ether}$ systems – contact ion pairs (CIPs)



K. A. Persson *et al.* *JACS* **2015**, *137*, 3411–3420

- Easy C-S bond dissociation in transient $[\text{Mg}^+ \text{TFSI}^-]$ CIPs

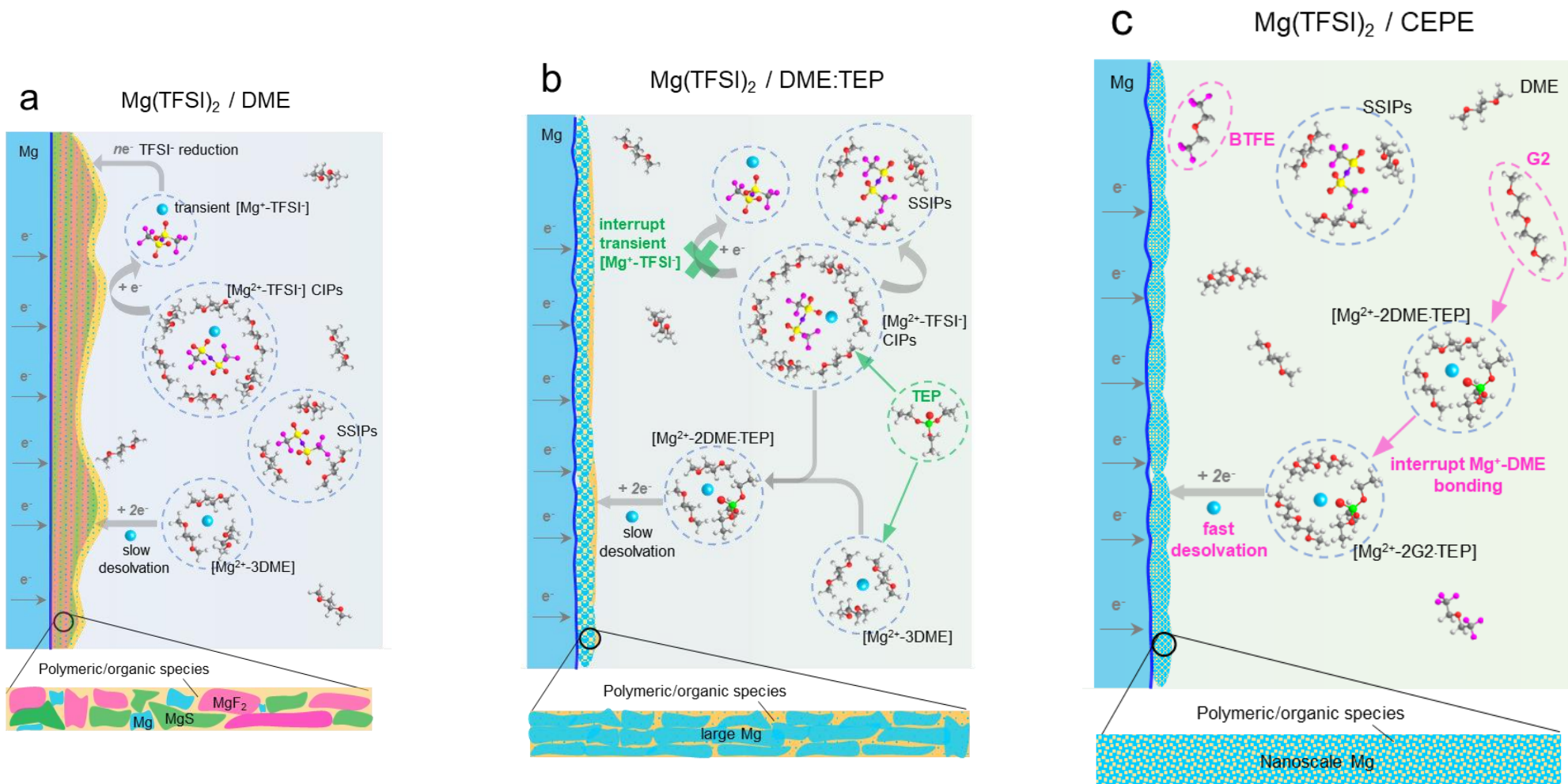


A. Baskin *et al.* *J. Phys. Chem. C* **2016**, *120*, 3583–3594

- Both anion and solvent molecules (G2) are also prone to reduction in highly coordinated CIPs

A Weakly Ion Pairing Electrolyte Designed for High Voltage Magnesium Batteries

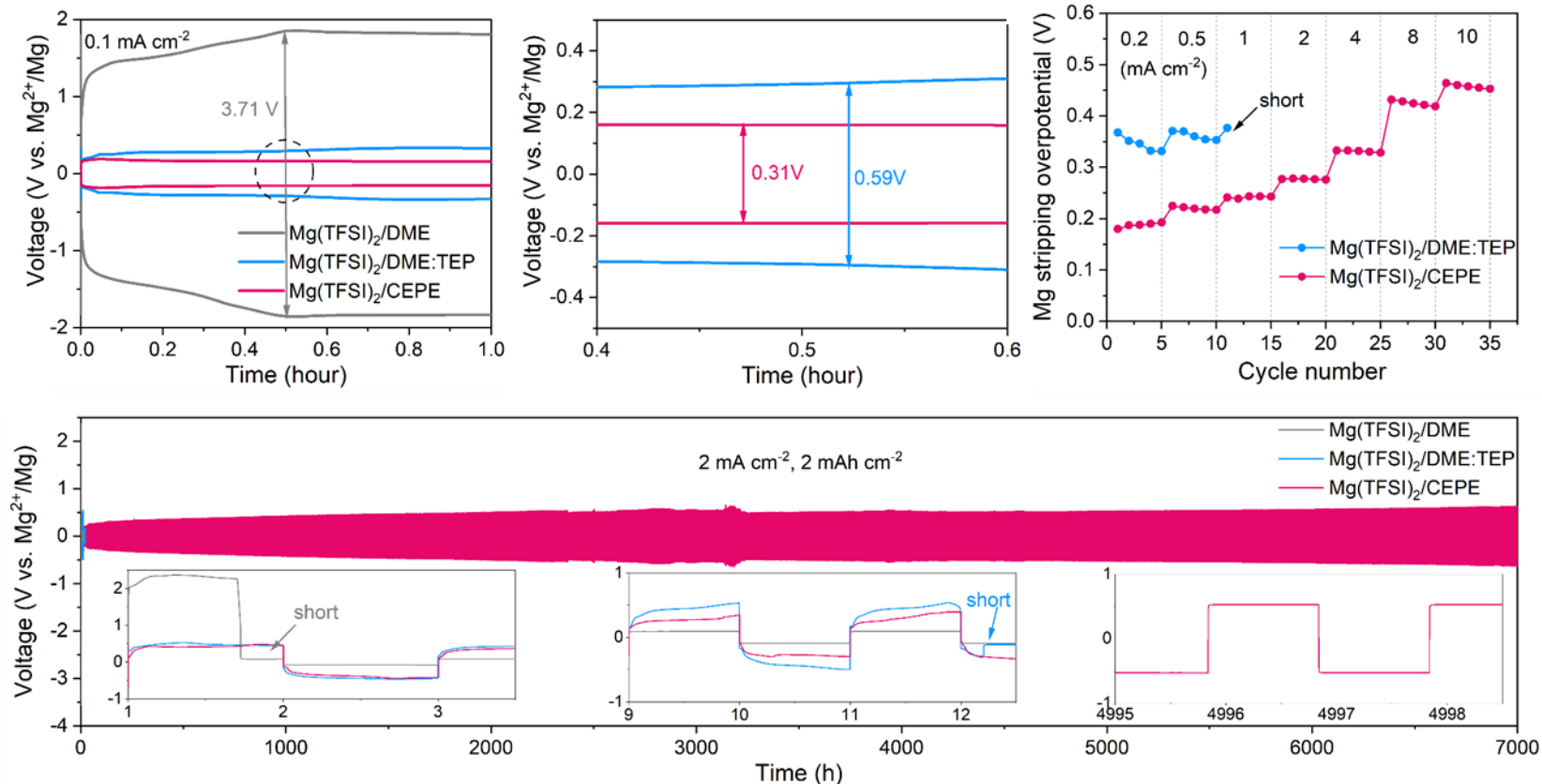
- Solvent-designed strategy disrupts ion pairing without need for costly/bulky anion salts



- CEPE:** Co-Ethereal Phosphate Electrolyte (solvent mixture of multiple ethers and phosphate)

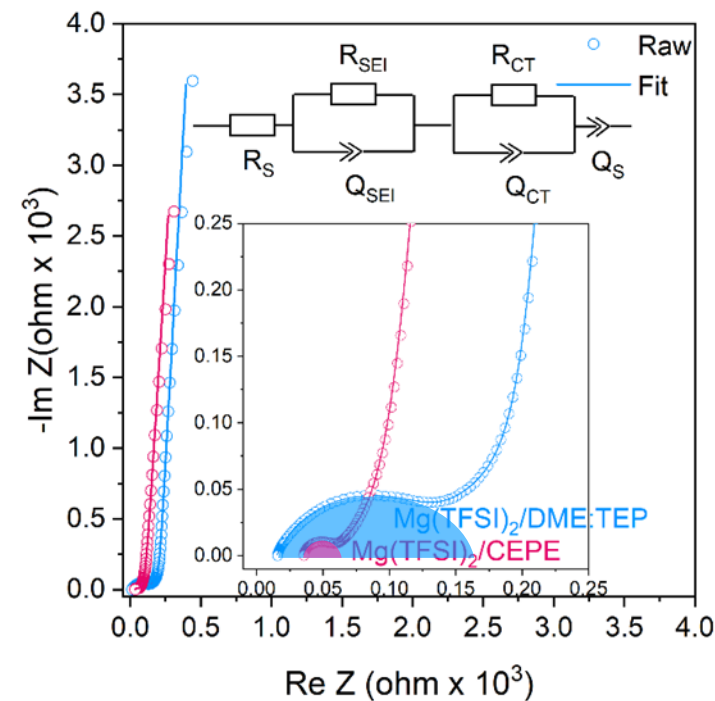
Highly reversible and stable Mg plating/stripping in CEPE

- Synergetic effect of phosphate and co-ethers



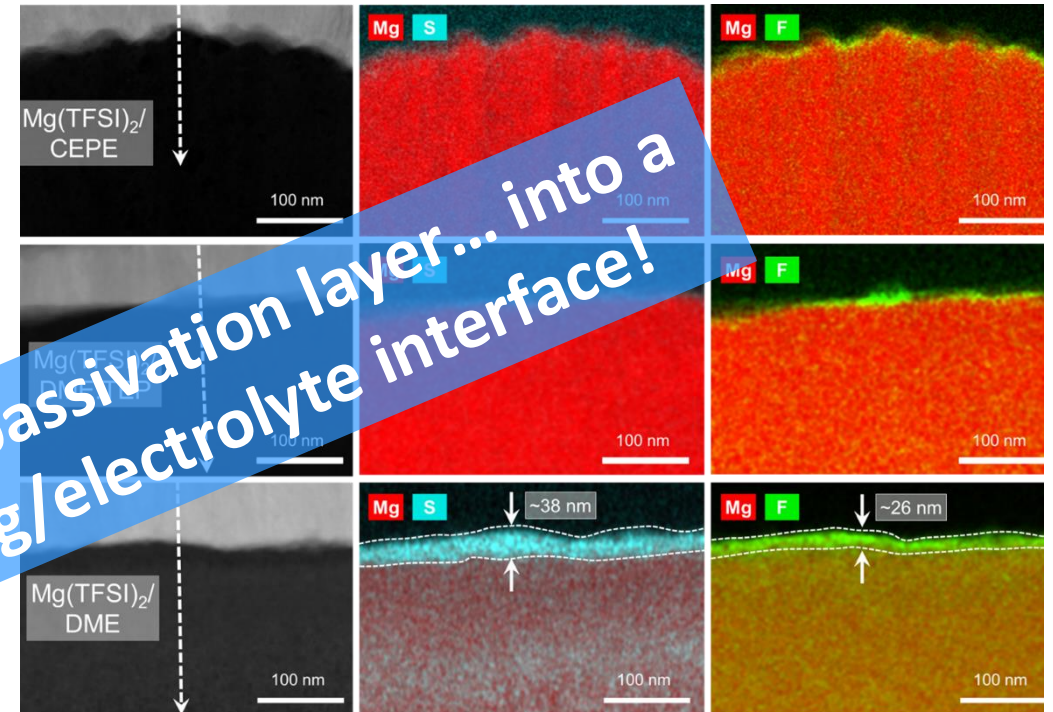
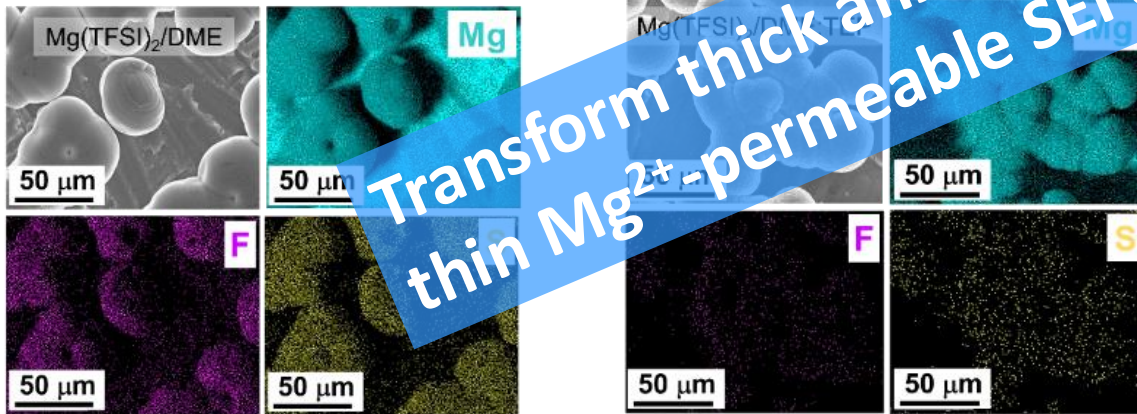
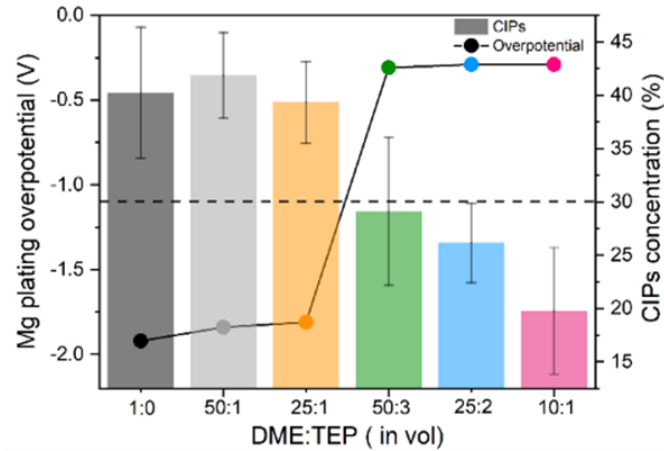
+ 95% Coulombic efficiency

Charge-transfer resistance at interface much less for CEPE

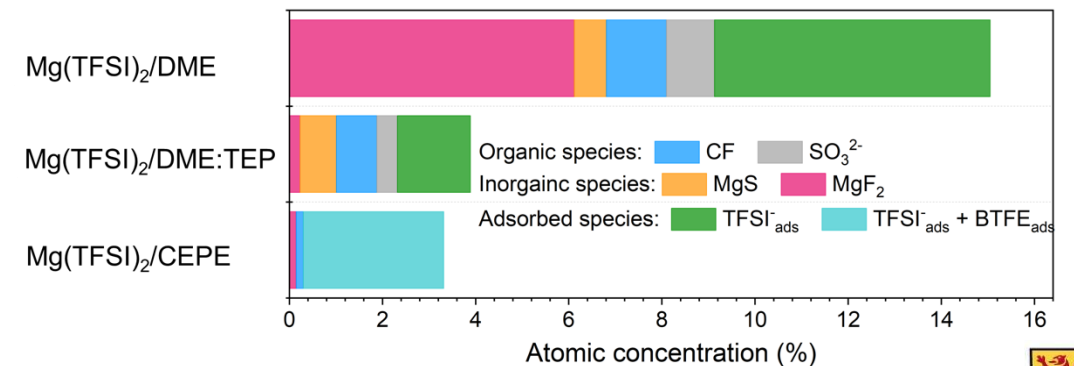


Minimized anion activity with phosphates and interfacial SEI chemistry

- Phosphate solvents effectively dissociate $[\text{Mg}^{2+}\text{-TFSI}^-]$ CIPs

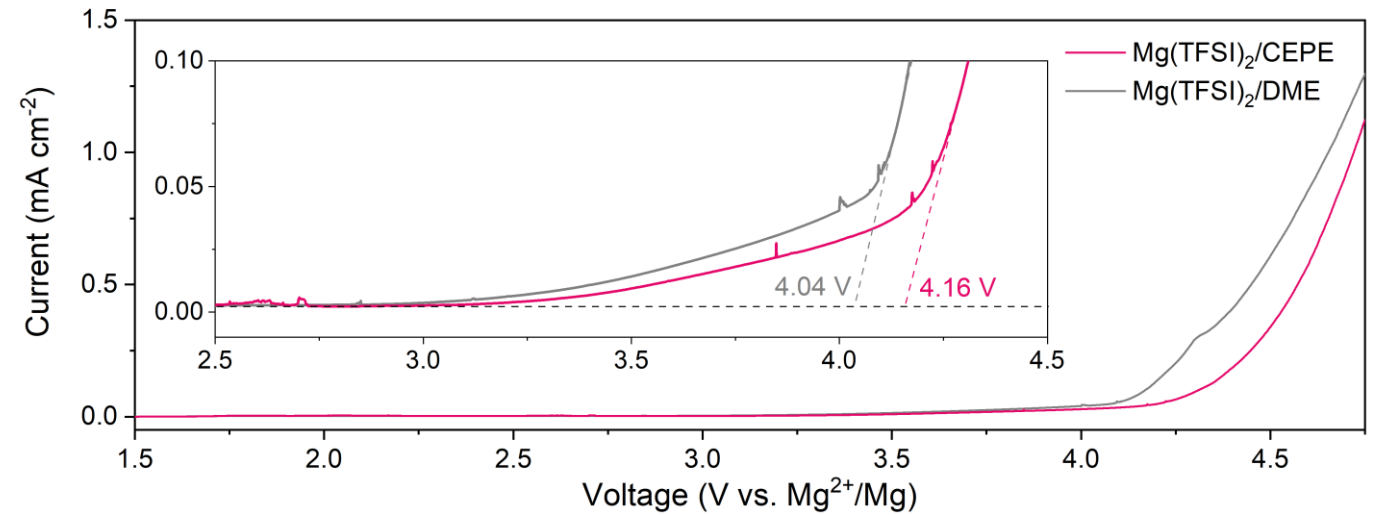
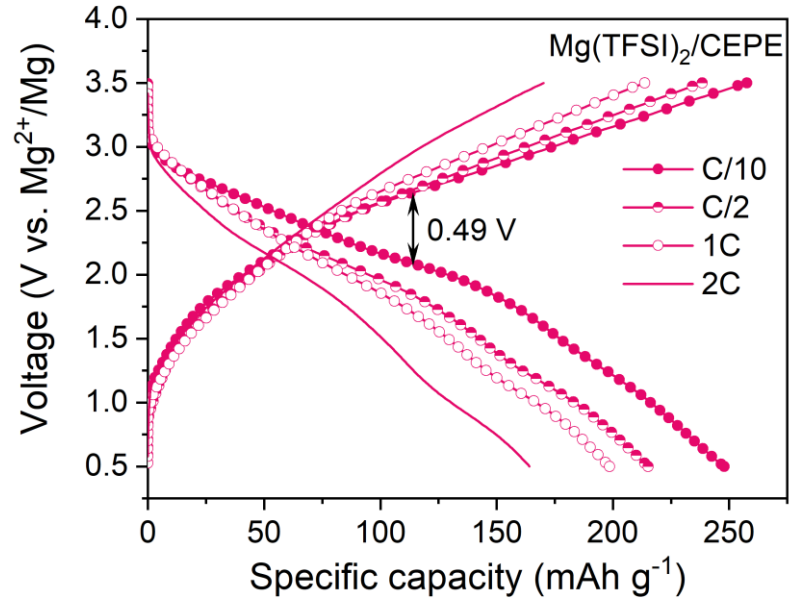


Transform thick anion-derived passivation layer... into a thin Mg^{2+} -permeable SEI on Mg/electrolyte interface!



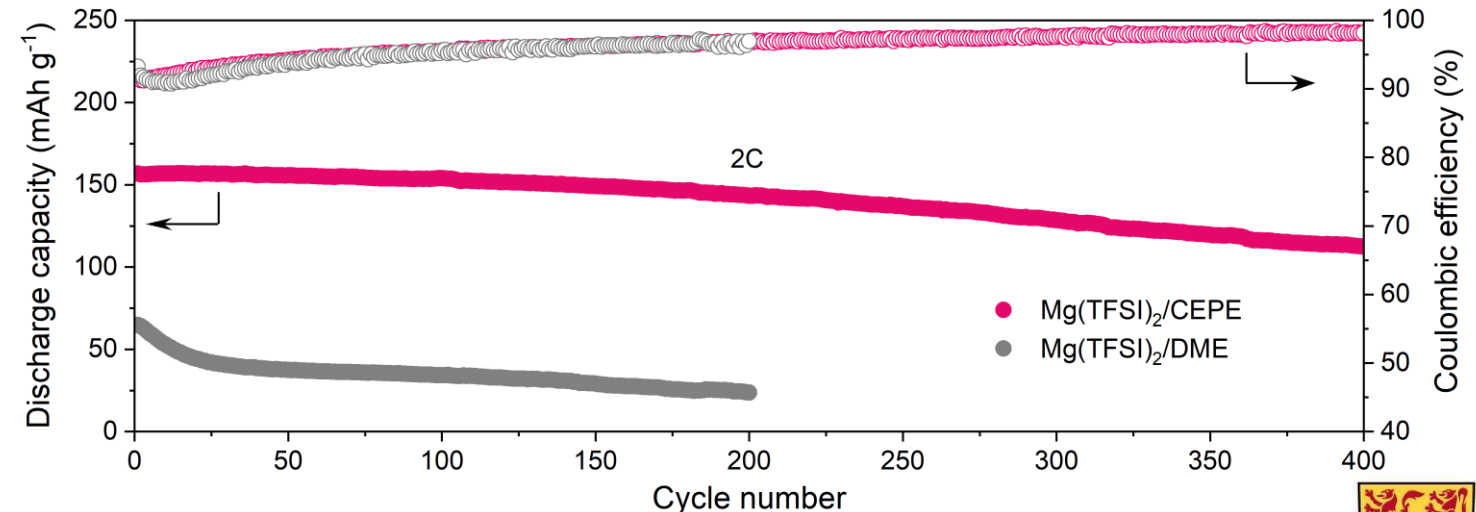
High-voltage stability in Mg-organic full cells

■ PANI as anion-intercalative type cathode



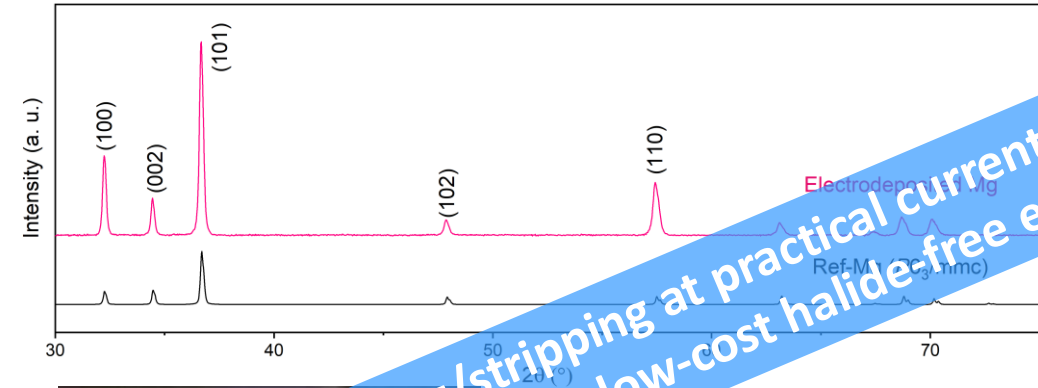
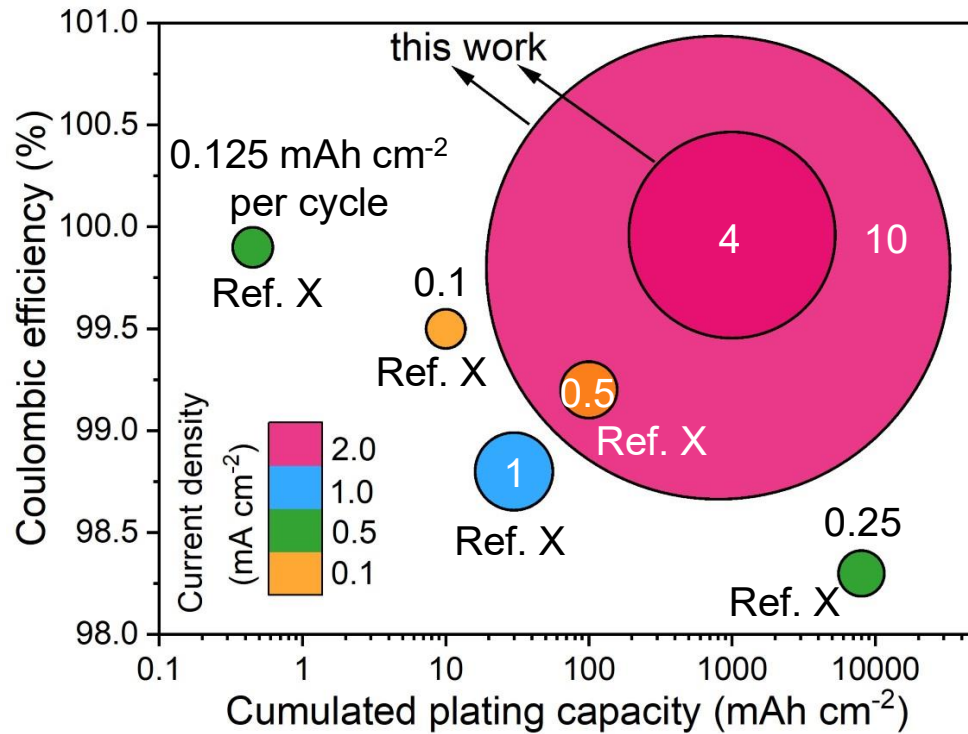
➤ Suppressed CIP formation simultaneously assists:

- Fast plating/stripping at the anode
- Facile TFSI⁻ intercalation at the cathode

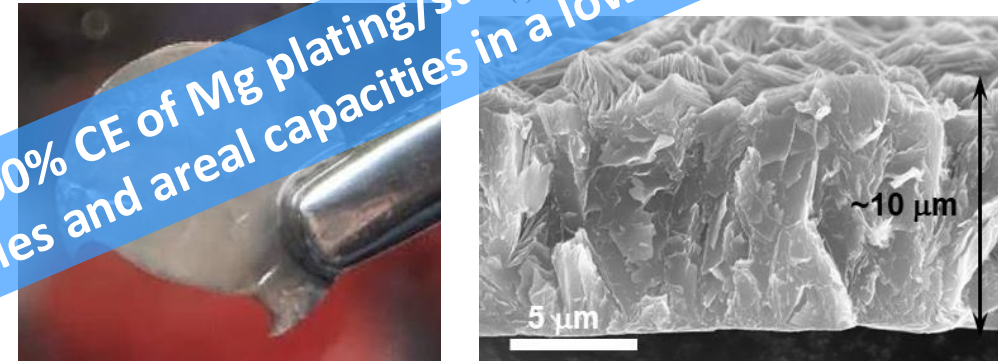


Achieving ~100% CE for Mg plating/stripping under demanding conditions

- A new generation of electrolyte (in progress) with tailored Mg^{2+} solvation structure



Near 100% CE of Mg plating/stripping at practical current densities and areal capacities in a low-cost halide-free electrolyte



Electrodeposited Mg foil peeled off Cu: 10 μm thick; 4 mAh cm^{-2}

Acknowledgements

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